

Operations Management

Session 2: Process Flow Management & Optimal Resource Allocation

Outline:

- Bottleneck and Cycle Time Analysis
 - Throughput versus Cycle Time Tradeoff
 - Balancing the line
 - Critical Path
 - Inventory Build-Up Charts
- National Cranberry
 - Process improvement
 - Capacity expansion analysis
- Optimal Resource Allocation
 - Merton's optimal production plan
 - Assessing the marginal value of resources

When Journalistic Intuition Violates Little's Law

Slower Lane

McDonald's customers are waiting longer at the drive-through according to QSR Magazine.

2004	Average time in seconds [☆]		2005	Average time in seconds [☆]
1. Wendy's	124.7		1. Wendy's	135.7
2. Checker's	131.5		2. Checker's	136.0
3. Taco Bell	148.2		3. Taco Bell	153.4
4. McDonald's	152.5		4. Burger King	166.1
5. Chick-fil-A	163.7		5. Rally's	167.0
6. Rally's	169.1		6. McDonald's	167.9

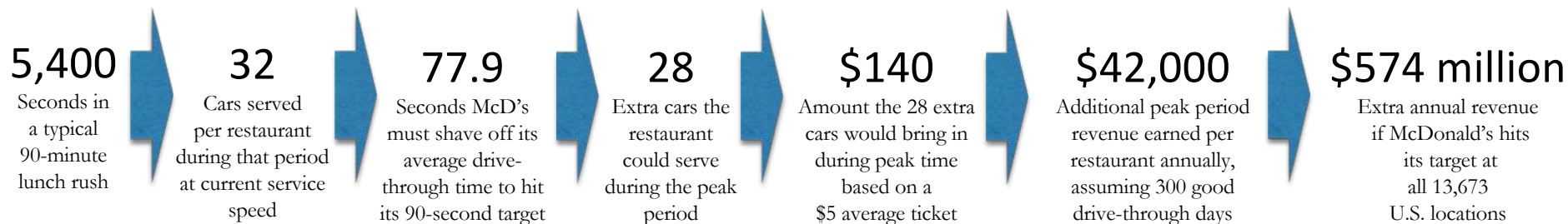
[☆] From the order point (menu board) to pickup-window handoff.



What would be the impact of cutting Avg. Time to 90 seconds?

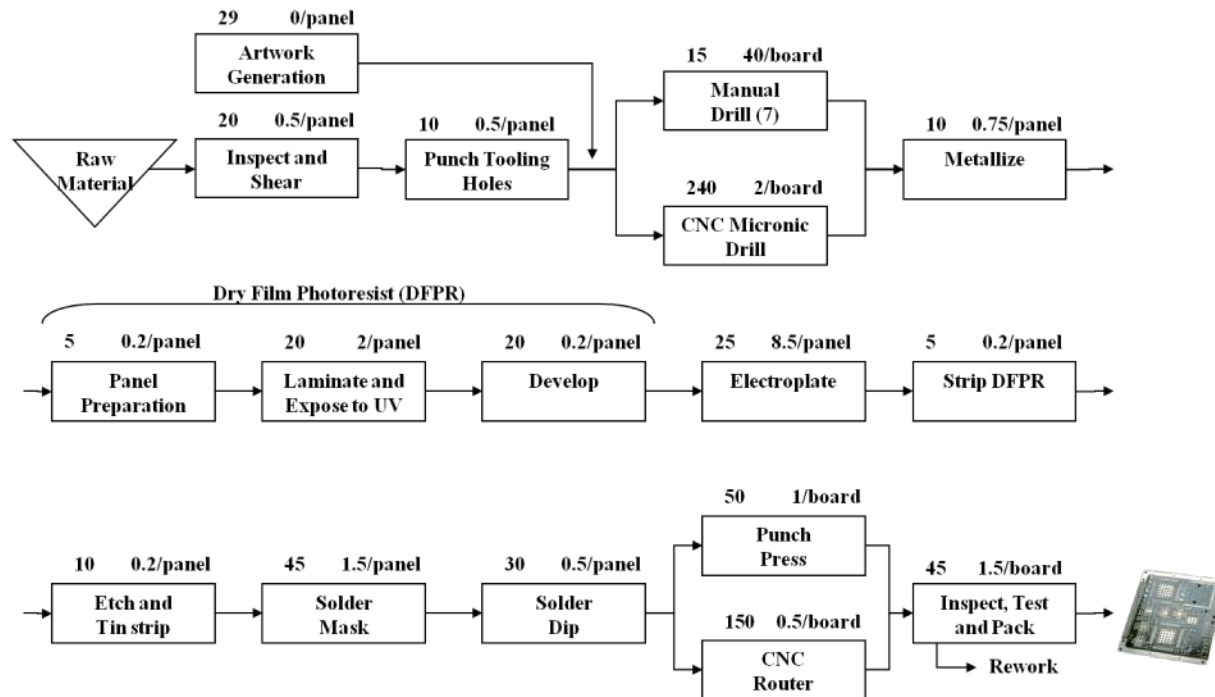
Time is Money

Drive-through service matters most during lunch, the peak time when fast food chain make 35% to 40% of their sales. McDonald's sluggish service is costing them – big time.



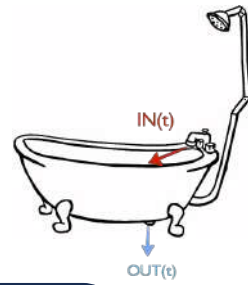
Process Flow Management

- How to measure performance (capacity, cycle time, inventory) for a network of activities, buffers and interfaces.
- How to design and manage a process to improve its overall performance.

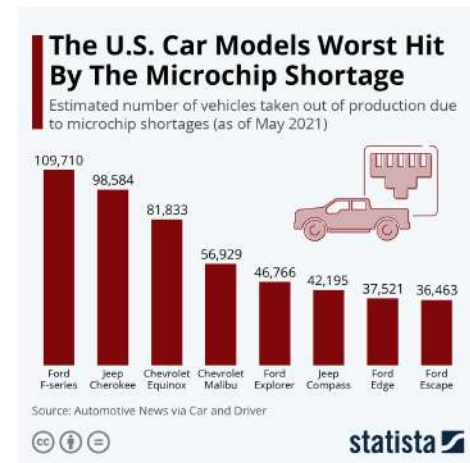
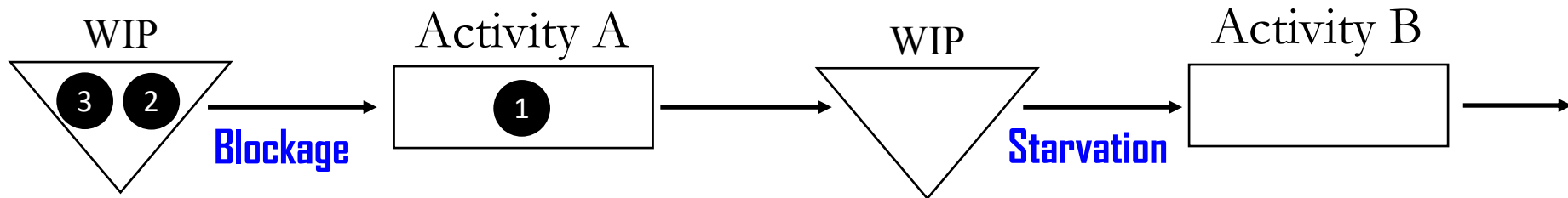


“Divide et impera.”
 (“Divide and conquer.”)
— *Attributed to Julius Caesar*

Key Concept: Process Balancing



Effective process flow management is about designing and managing processes (activities and their interfaces) to remove:
BLOCKAGE and STARVATION.



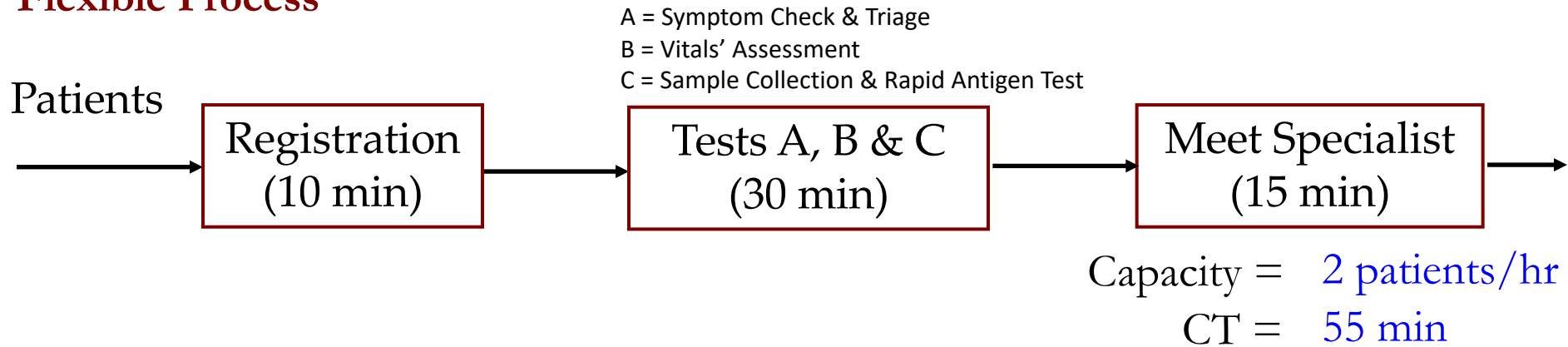
Key Concept: Capacity Bottleneck

- A resource or resource pool that limits the maximum average sustainable throughput
- The bottleneck is the system's pacemaker:
 - Determines system's capacity and long-run output levels
 - Work-in-process (WIP) accumulates in front of the bottleneck
- Protecting the bottleneck is critical
 - Variability, downtime, and rework at the bottleneck directly reduce system throughput
 - One lost unit of bottleneck time is one lost unit of system output

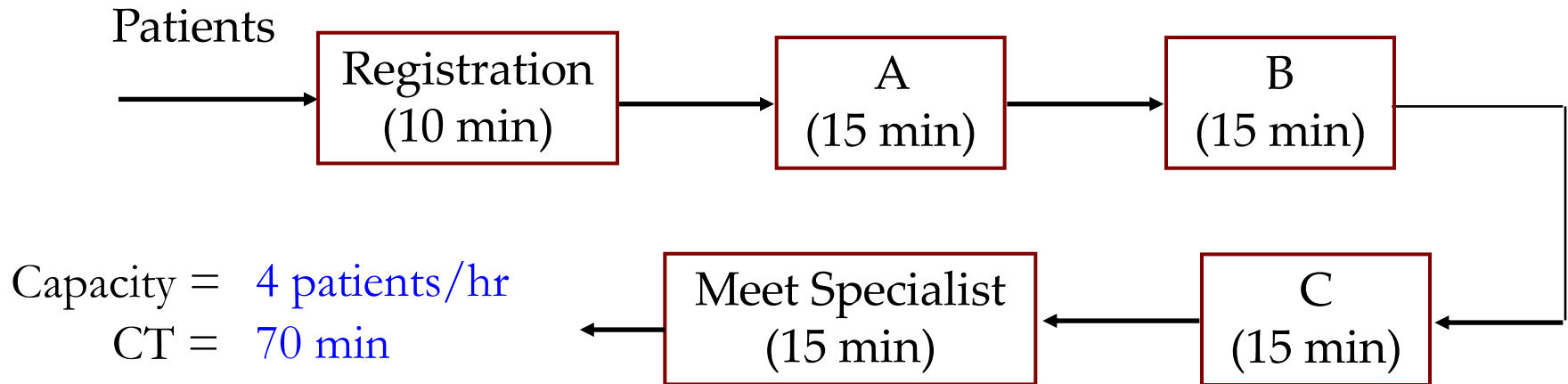


Process Flow Analysis: Example I

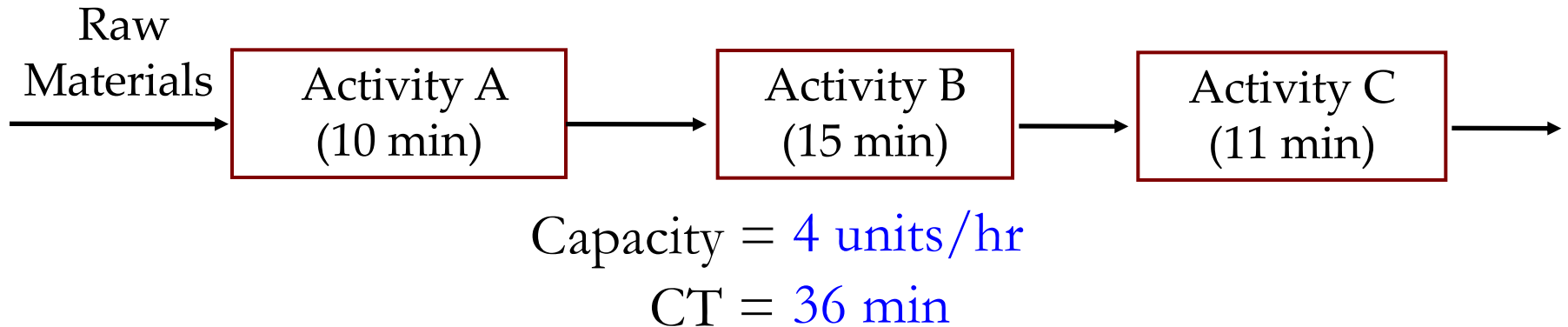
Flexible Process



Dedicated Process

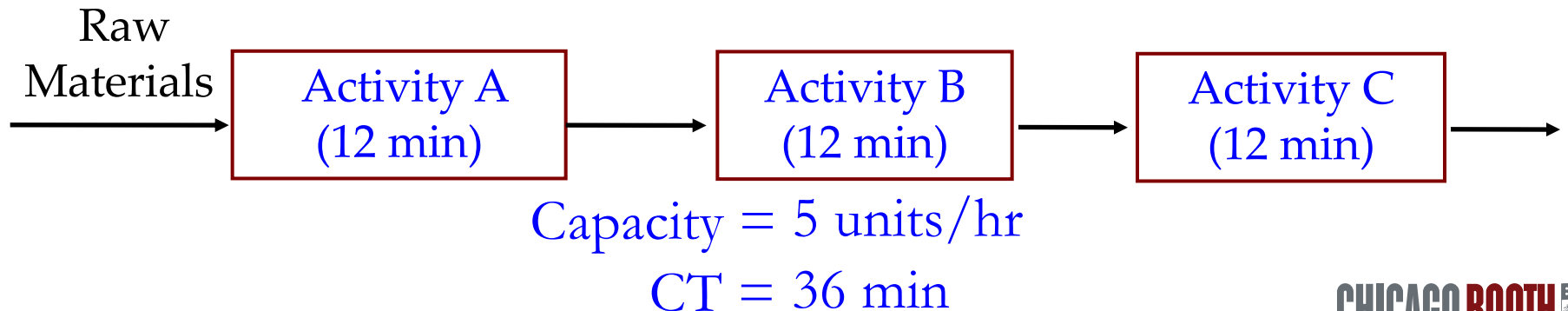


Process Flow Analysis: Example II



How can you improve this process?

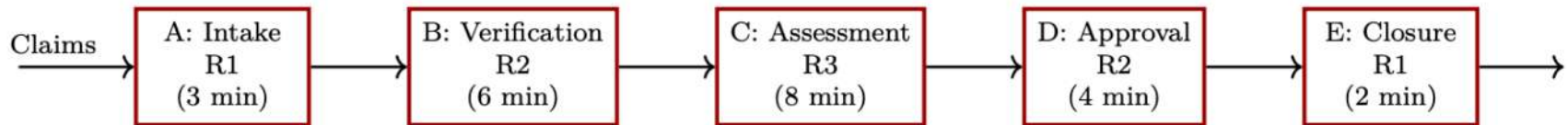
If work can be redistributed among activities, then there is an opportunity to do “Line Balancing”



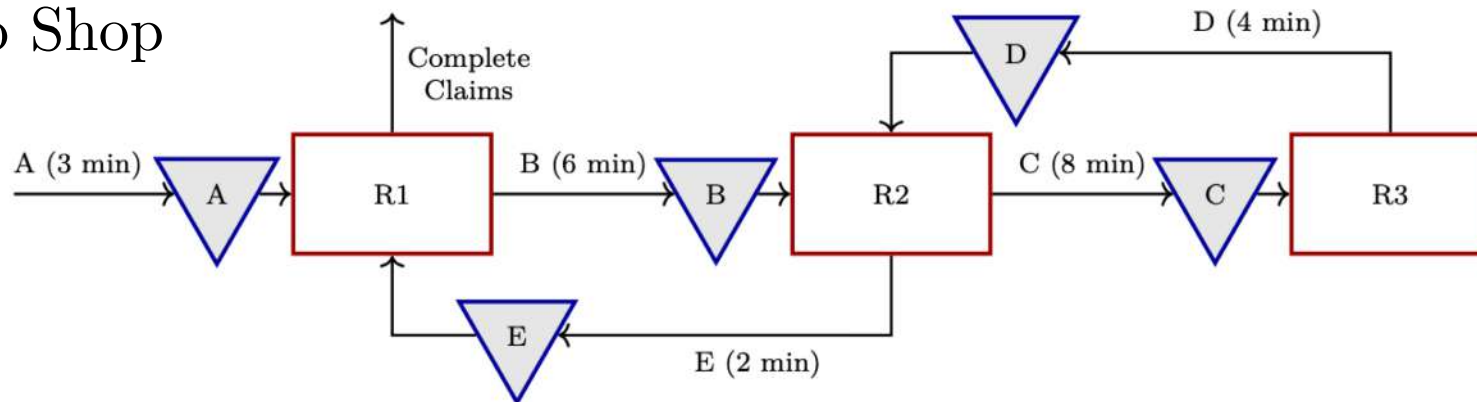
Process Flow Analysis: Example III

- A. Intake (create case record)
- B. Verification (check policy and coverage)
- C. Assessment (estimate loss)
- D. Approval (authorize payment)
- E. Closure (notify and archive)

R1: Clerk
R2: Analyst
R3: Supervisor



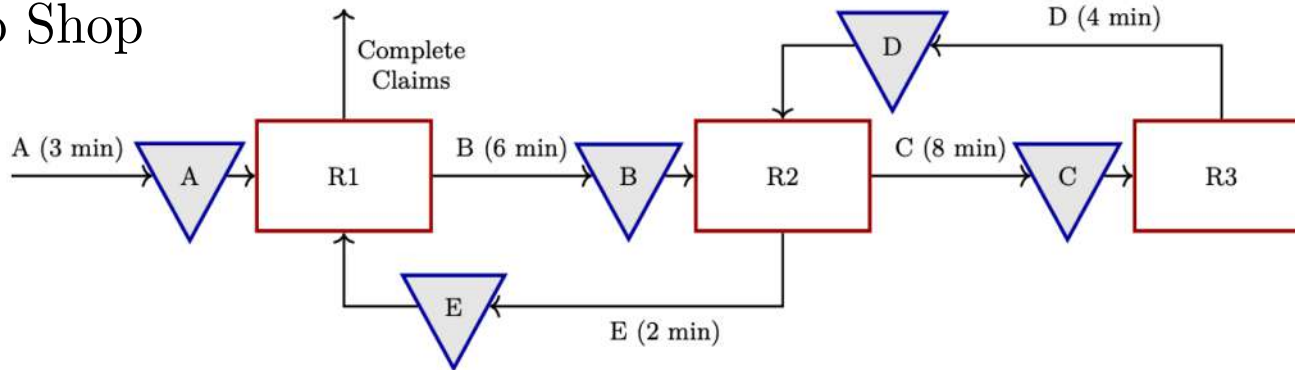
Job Shop



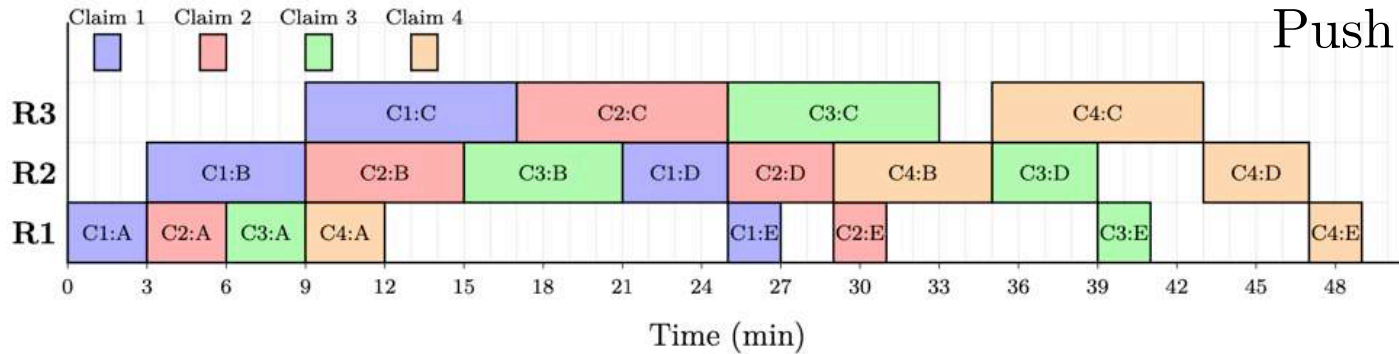
$$\begin{aligned} \text{R1 load/job} &= 3 + 2 = 5 \text{ min} \Rightarrow \text{R1 capacity} = 60/5 = 12 \text{ jobs/hr,} \\ \text{R2 load/job} &= 6 + 4 = 10 \text{ min} \Rightarrow \text{R2 capacity} = 60/10 = 6 \text{ jobs/hr,} \\ \text{R3 load/job} &= 8 \text{ min} \Rightarrow \text{R3 capacity} = 60/8 = 7.5 \text{ jobs/hr.} \end{aligned}$$

Work Scheduling: Gantt Charts

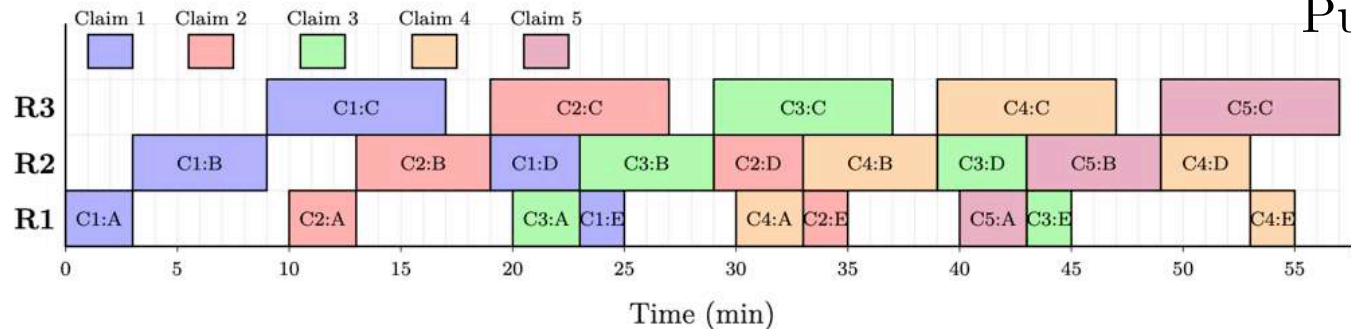
Job Shop



Push System



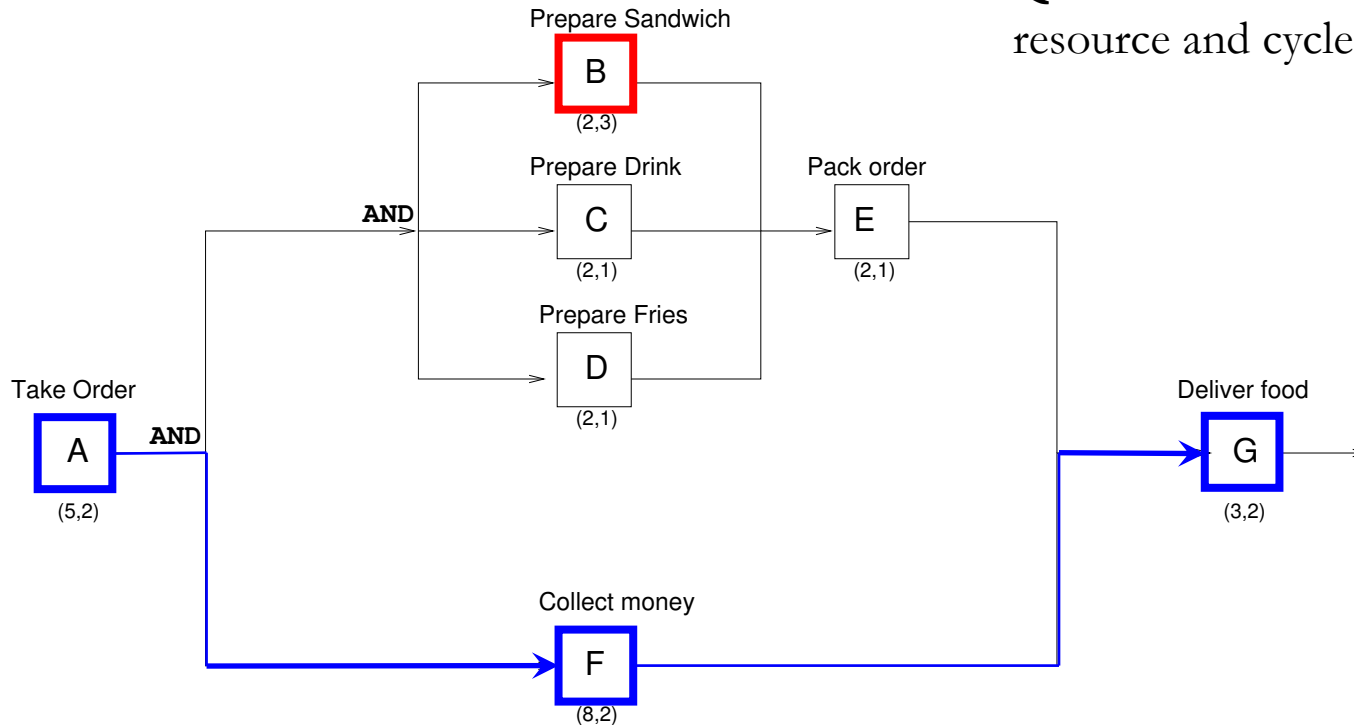
Pull System



Process Flow Analysis: Example IV

Drive Through Window ---- Fast Food Process Flow

Question: What are the bottleneck resource and cycle time of the system?



Path	Length
ABEG	20
ACEG	18
ADEG	18
AFG	22

KEY: (Average wait time, Average processing time)
All times in minutes

The **critical path** is the longest path through a process which all jobs must pass.

Whereas **capacity** is determined by a **bottleneck**, **cycle-time** is determined by a **critical path**.

National Cranberry Cooperative



Available in YouTube [[Link](#)]

NCC: Overview



➤ What are the Main Operational Problems?

- Symptoms
- Causes

➤ What would you do to improve the Operation?

Process Flow Diagram

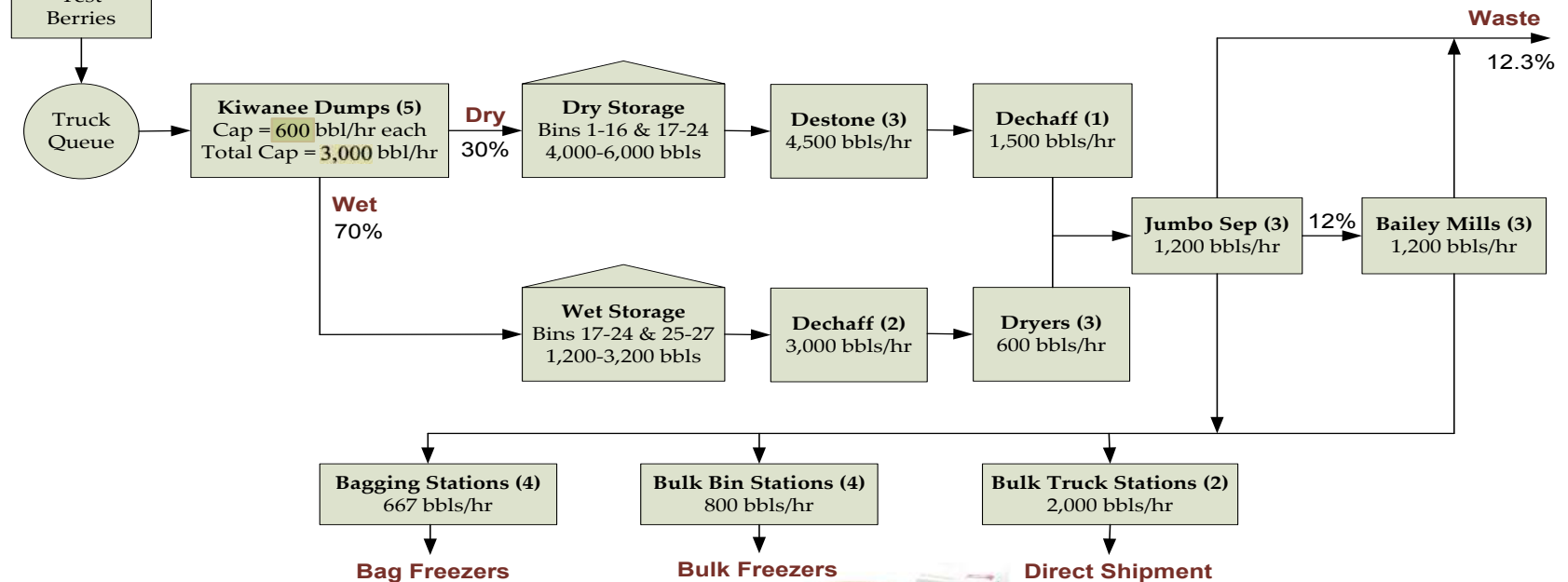


Working Conditions - Input (Peak Day):

Total = 18,000 bbls per day

Dry (30%) = 5,400 bbls per day = 450 bbls per hour (7am to 7pm)

Wet (70%) = 12,600 bbls per day = 1,050 bbls per hour



Bottlenecks?



NCC: Load Factor Analysis

Consider the following operating conditions:

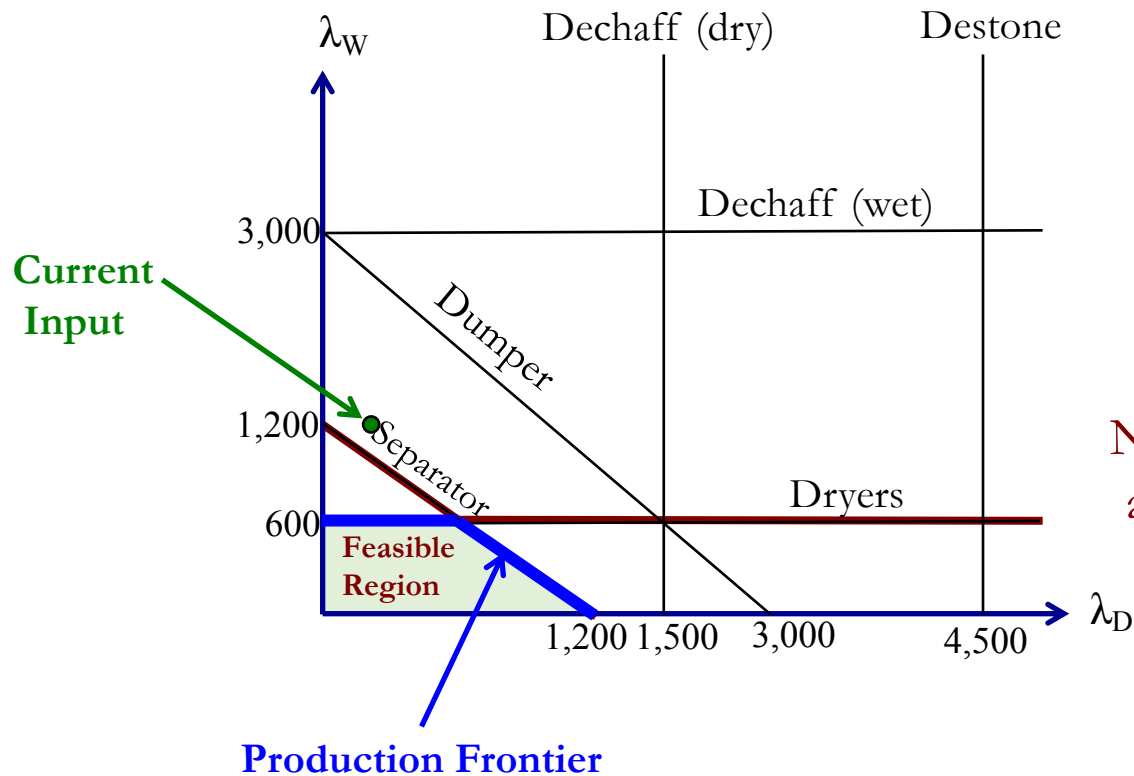
Operating Conditions		
Normal operating hours	12	hrs
Total demand	18,000	bbls
Fraction of wet berries	70%	
Fraction of dry berries	30%	

Load Factor Analysis

Activity	Capacity				Demand			Load Factor	Required Processing Time
	# units (-)	Capacity per unit (bbls/hr)	Time Availability (hrs)	Available Capacity bbls	Wet bbls	Dry bbls	Total bbls		
Dumpers	5	600	12	36000	12600	5400	18000	50.0%	6.0
Destone	3	1500	12	54000	0	5400	5400	10.0%	1.2
Dechaff (dry)	1	1500	12	18000	0	5400	5400	30.0%	3.6
Dechaff (wet)	2	1500	12	36000	12600	0	12600	35.0%	4.2
Dryers	3	200	12	7200	12600	0	12600	175.0%	21.0
Jumbo Sep	3	400	12	14400	12600	5400	18000	125.0%	15.0
Bailey Mills	3	1200	12	43200			2160	5.0%	0.6
Bagging	4	167	12	8016			5262	65.6%	7.9
Bulk Bin	4	200	12	9600			5262	54.8%	6.6
Bulk Truck	2	1000	12	24000			5262	21.9%	2.6

NCC: Production Frontier

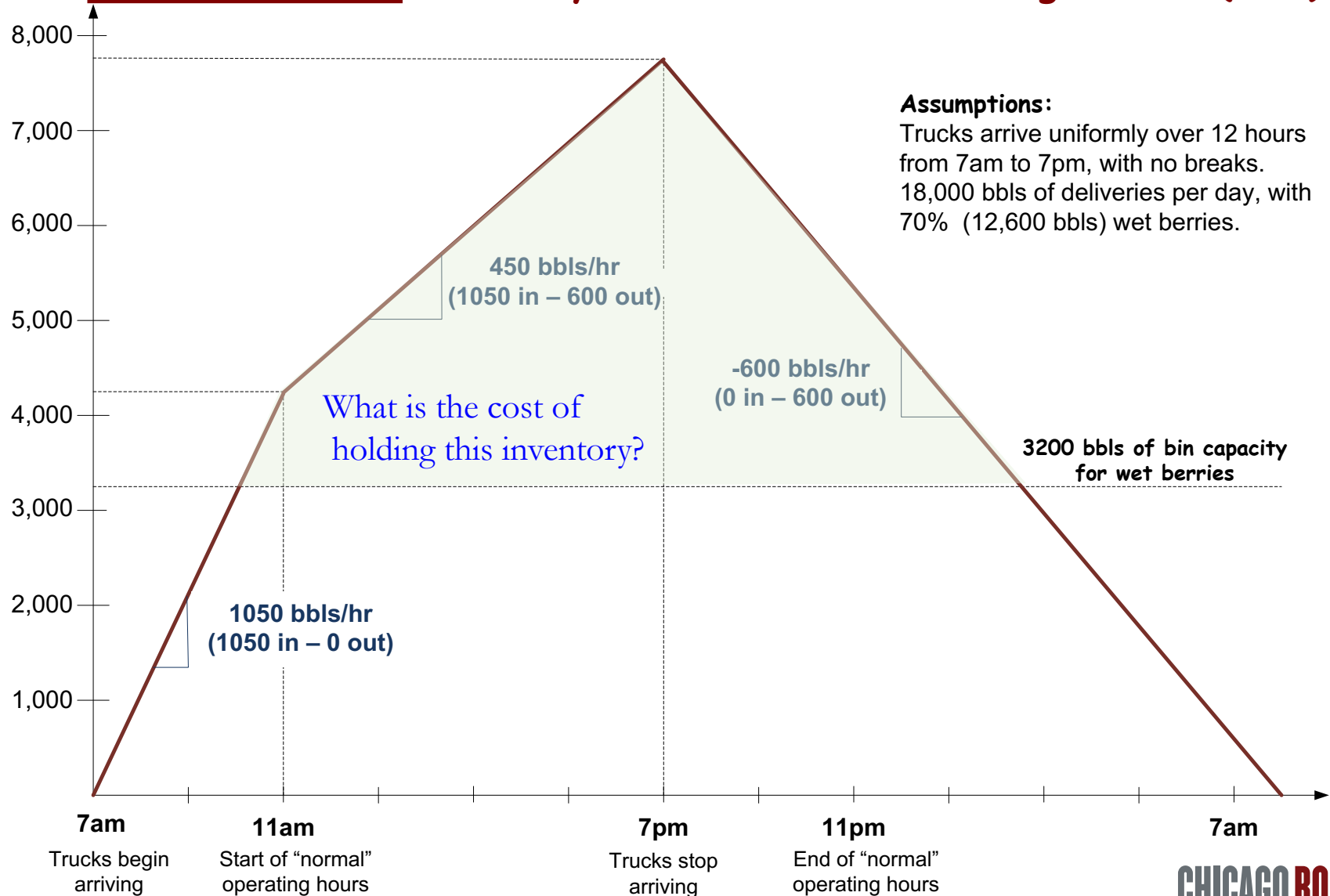
Question: How production capacity changes as a function of the input of dry (λ_D) and wet (λ_W) berries? (Assume λ_W and λ_D are measured in **bbls/hr**)



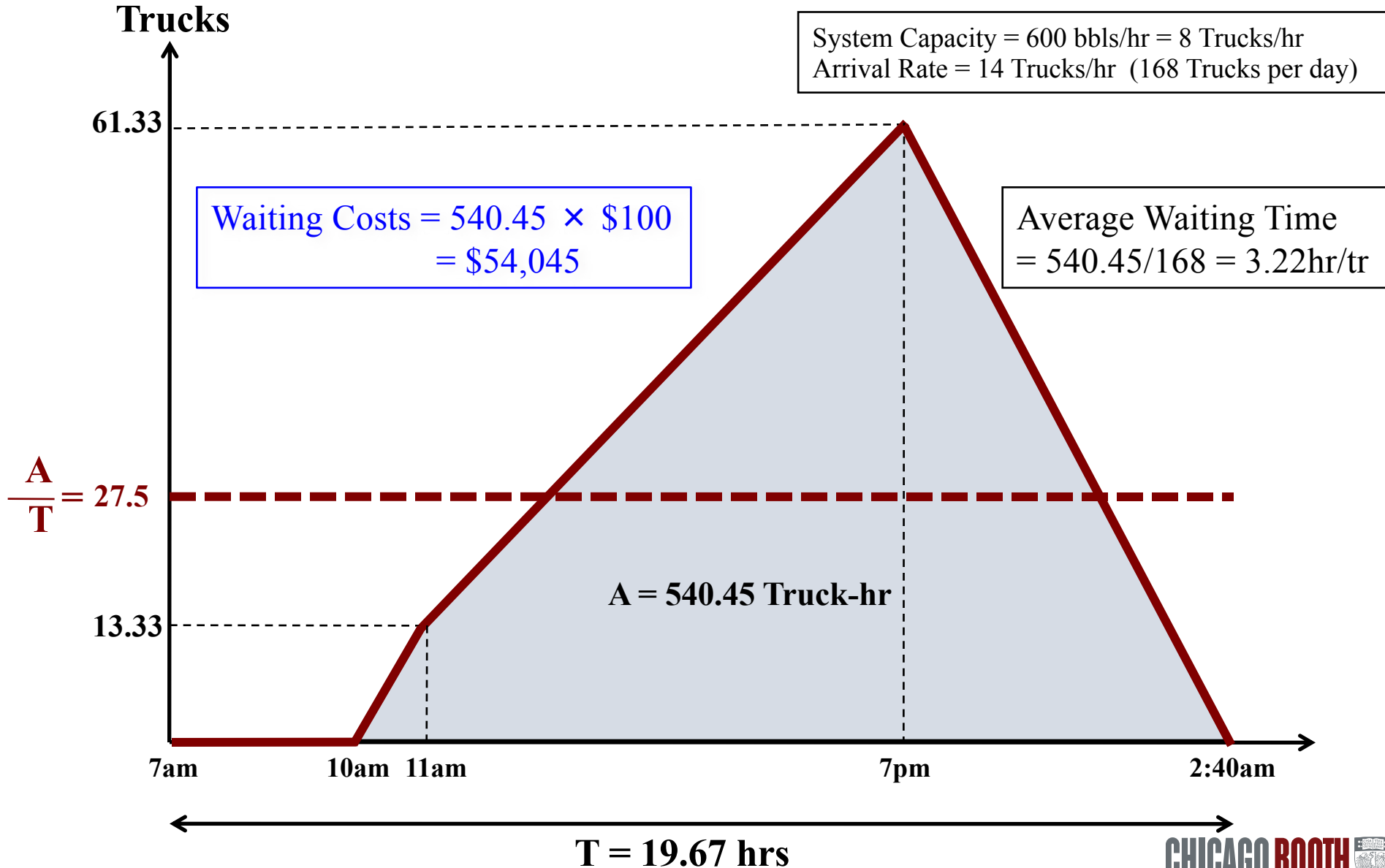
NCC needs more Dryer
and Separator capacity
Overtime

NCC: Inventory Build-Up Chart

Wet Cranberries in the System in Bins and Waiting Trucks (bbls)



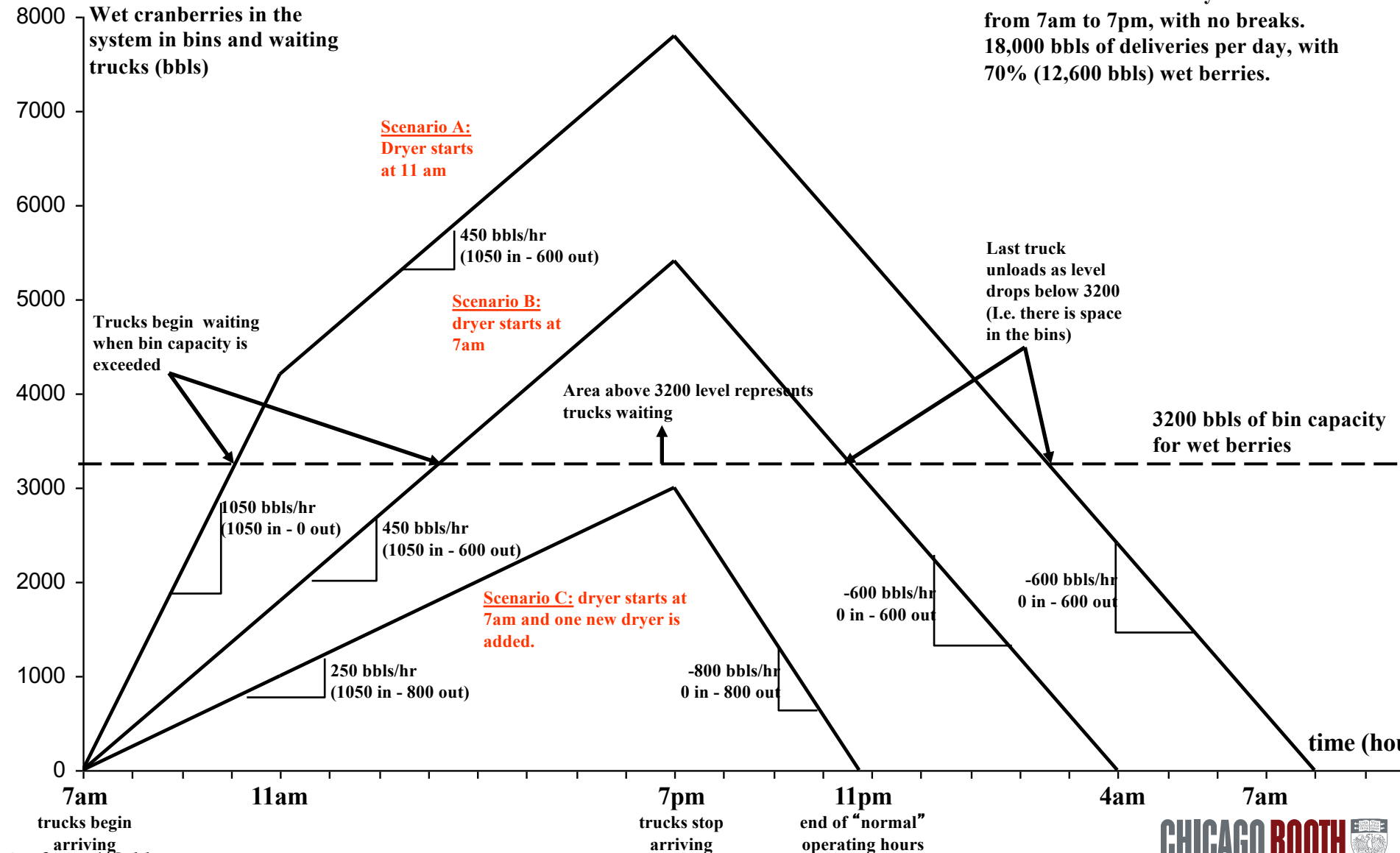
Truck Waiting Time



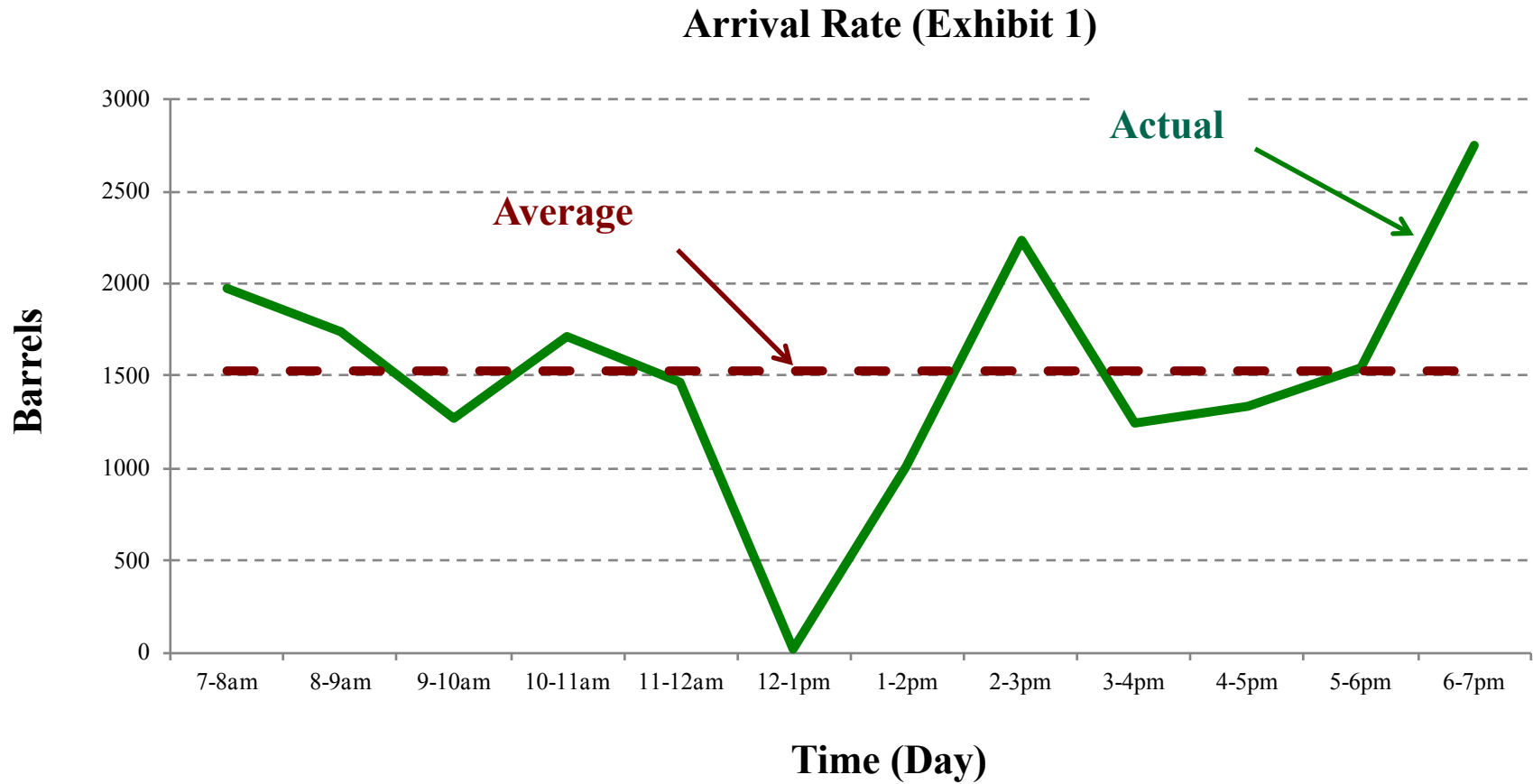
Plot of cranberries in bins and trucks over the course of the day under 3 alternative scenarios

Assumptions:

Trucks arrive uniformly over 12 hours
from 7am to 7pm, with no breaks.
18,000 bbls of deliveries per day, with
70% (12,600 bbls) wet berries.

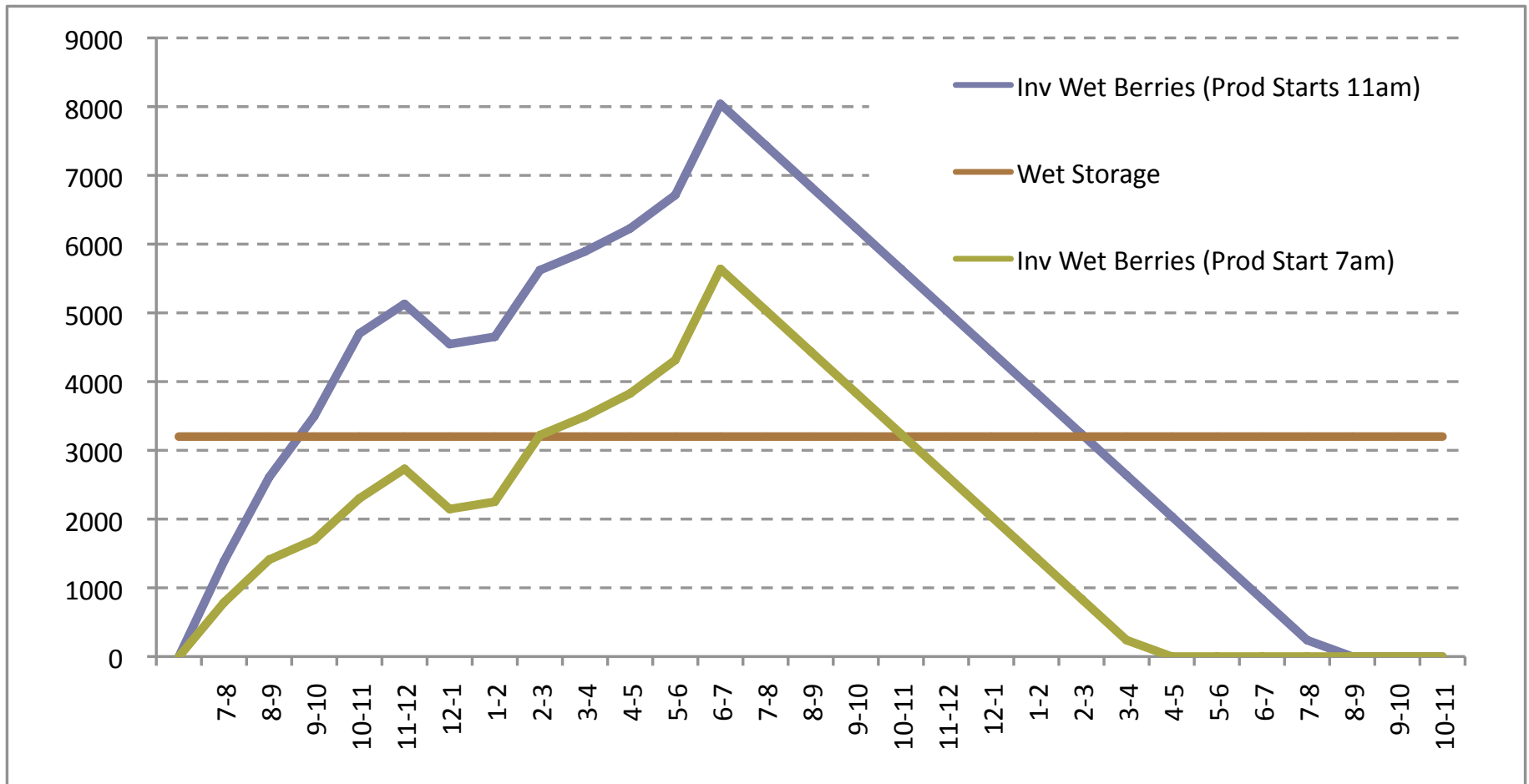


Arrival Pattern



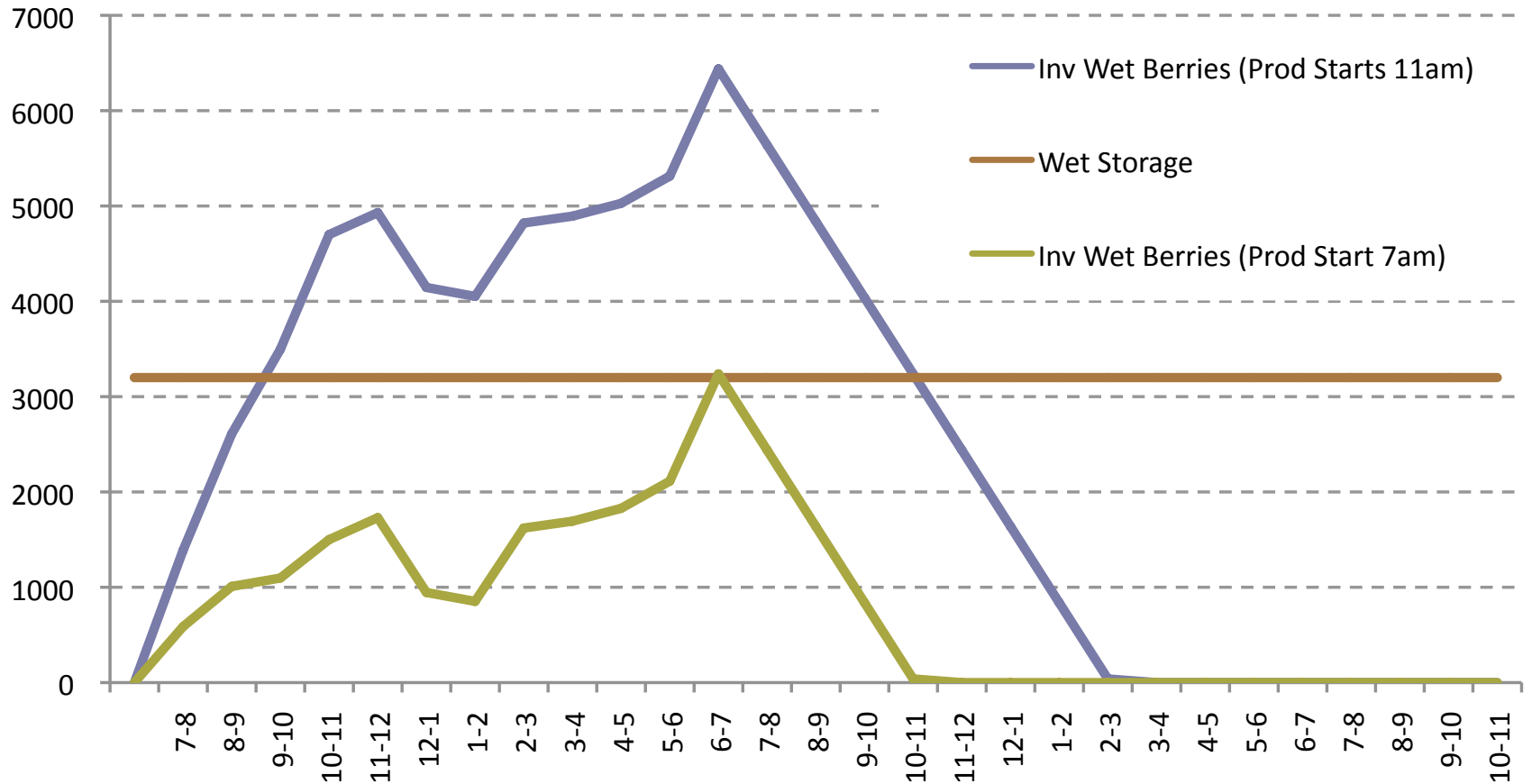
NCC: Impact of the Arrival Pattern

Current Configuration



NCC: Impact of the Arrival Pattern

Adding One Additional Dryer



NCC: Effects of Production Variability

Production (bbls/day)	Frequency (N ^o Days)	Overtime (hr)	Truck Waiting (truck-hr)	ΔCost (\$)
10000<	80	0	0	\$0
11,000	1	0	0	\$0
12,000	0	0	0	\$0
13,000	2	0	0	\$0
14,000	2	0	0	\$0
15,000	2	2	0.4	\$256
16,000	4	3	19.9	\$2,315
17,000	3	4	63.3	\$6,762
18,000	6	5	125.5	\$13,088

Season Cumulative ΔCost \$108,587

Assumptions:

- No Additional Dryer
- 70% wet berries
- Production starts at 7am
- Nine workers during overtime @ \$12/hr

Optimal Resource Allocation

(Solving Linear Programs)

Many business problems involved the allocation of limited resources among a set of competing set of activities, while satisfying business rules

- Production Scheduling: allocate production capacity among product lines
- Portfolio Management: allocate funds among financial instruments
- Assortment Planning: allocate shelf space among alternative brands
- Logistics & Routing: allocate transportation capacity to distribute parcels

Activity-Resource Consumption Matrix							
	Activity 1	Activity 2		Activity j		Activity m	Capacity
Resource 1	a_{11}	a_{12}		a_{1j}		a_{1m}	b_1
Resource 2	a_{21}	a_{22}		a_{2j}		a_{2m}	b_2
....							
Resource i	a_{i1}	a_{i2}		a_{ij}		a_{im}	b_i
....							
Resource n	a_{n1}	a_{n2}		a_{nj}		a_{nm}	b_n
Contribution	c_1	c_2		c_j		c_m	

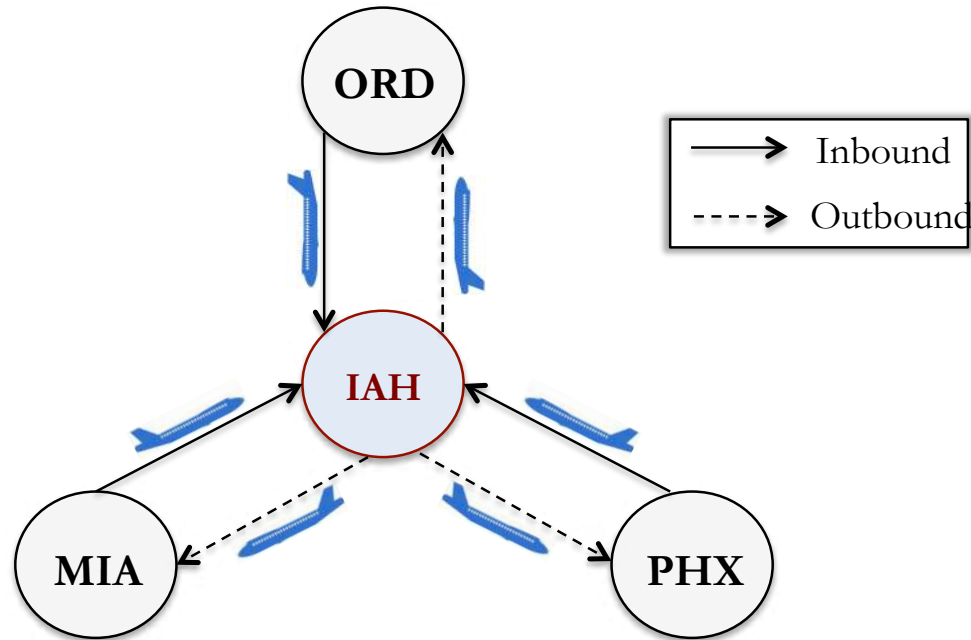
Transportation

From \ To	Columbus	St. Louis	Denver	Los Angeles	Factory supply
Indianapolis	10 25	5 35	36	60	15
Phoenix	55	6 30	25	25	6
New York	40	1 50	13 80	90	14
Atlanta	30	40	2 66	9 75	11
Destination requirements	10	12	15	9	46 46

Cost = \$2,502

Question: How many units to ship from each factory to each destination?

Airline Revenue Management



Six resources (flights) with capacity 240 seats each.

Price for each Passenger Route

Origin	Destination			
	Houston	Chicago	Miami	Phoenix
Houston	-	\$197	\$110	\$125
Chicago	\$190	-	\$282	\$195
Miami	\$108	\$292	-	\$238
Phoenix	\$110	\$192	\$230	-

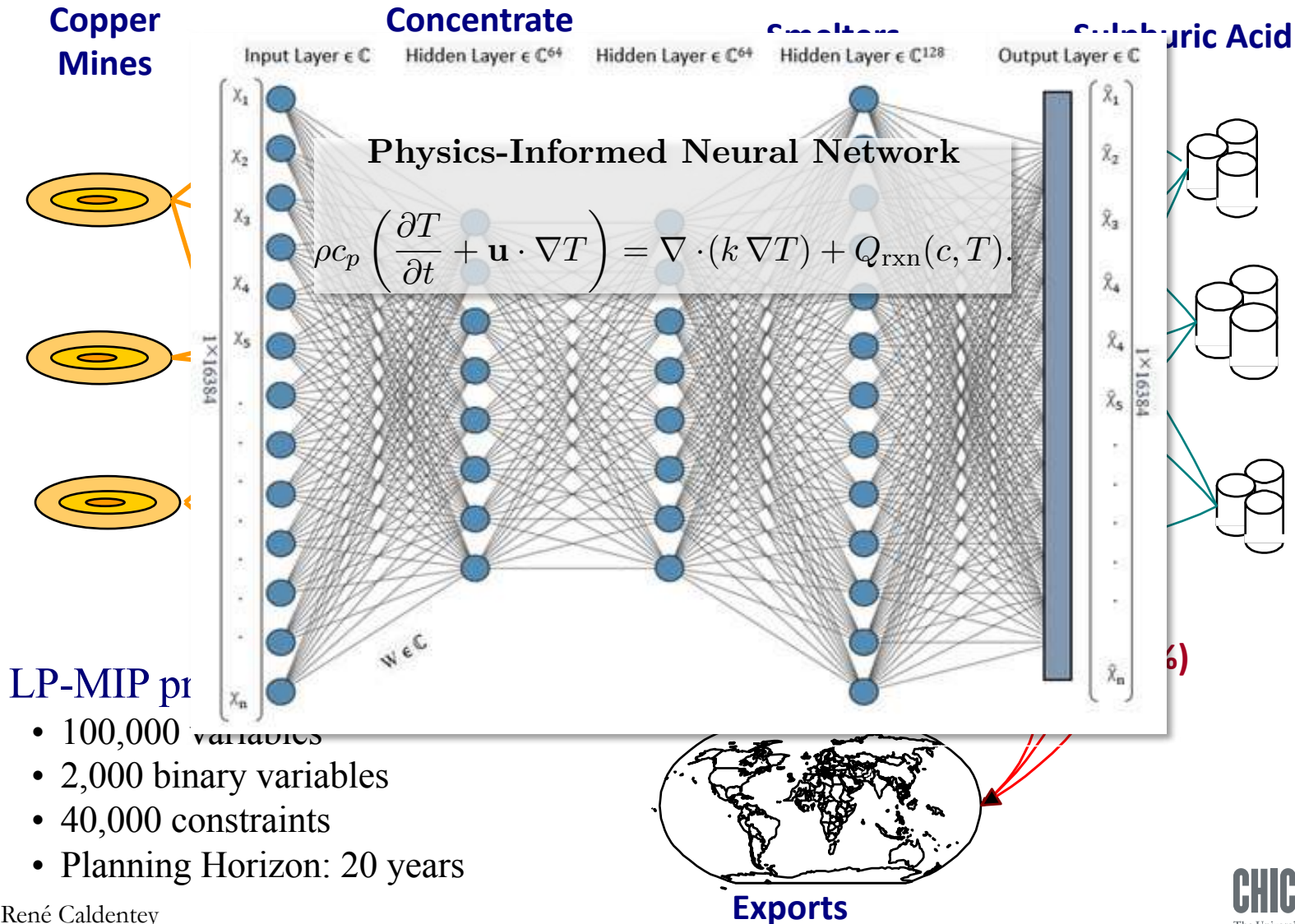
Demand During one Bank

Origin		Destination				Total Demand for flight to hub	
		Houston	Chicago	Miami	Phoenix		
	Houston	0	123	80	110	-	
	Chicago	130	0	98	88	316	(C-H)
	Miami	72	105	0	68	245	(M-H)
	Phoenix	115	90	66	0	271	(P-H)
Total demand for flight to spoke		-	318 (H-C)	244 (H-M)	266 (H-P)		

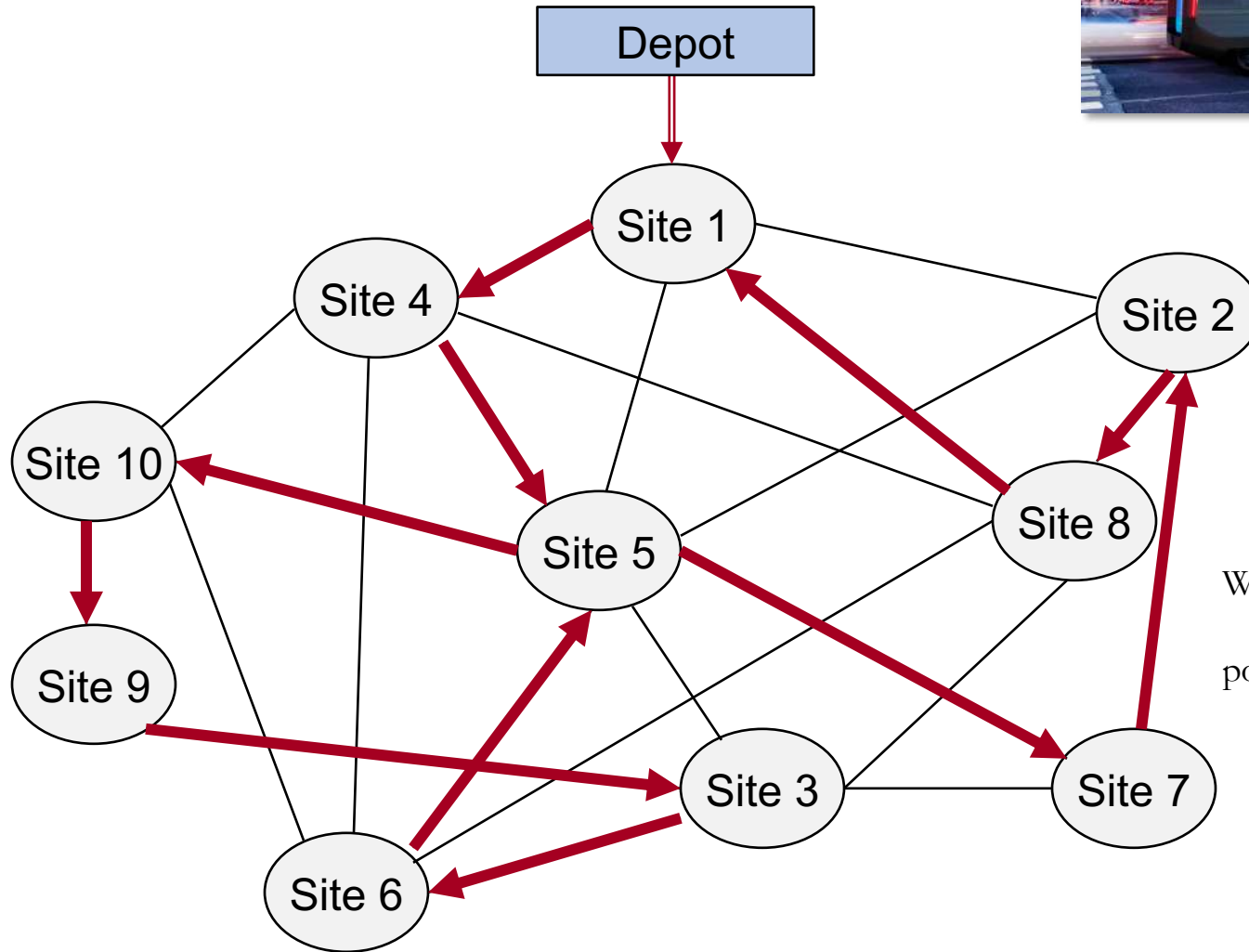
Problem: What is the optimal number of tickets to sell on each (O-D) route?

Production Process

Question: What is an optimal production plan?



Logistics and Distribution



With 10 sites there are
 $10! = 3,628,800$
possible tours

Problem: What is an optimal tour, i.e., a tour that minimizes the total distance?

Currency Opportunity



Symbol	 USD	 EUR	 GBP	 JPY	 CHF	 CAD	 AUD
 USD	1	0.7469	0.6416	77.0490	0.9078	1.0502	1.0351
 EUR	1.3388	1	0.8590	103.1510	1.2158	1.4062	1.3857
 GBP	1.5584	1.1644	1	120.0720	1.4153	1.6368	1.6131
 JPY	0.0130	0.0097	0.0083	1	0.0115	0.0136	0.0134
 CHF	1.1013	0.8226	0.7065	84.8450	1	1.1563	1.1399
 CAD	0.9529	0.7113	0.6116	73.3630	0.8646	1	0.9858
 AUD	0.9662	0.7218	0.6198	74.4450	0.8771	1.0149	1

Question: Is there any currency arbitrage opportunity (free lunch)?

Merton Truck Company



What is the main issue facing Merton Truck?

What product mix to select to maximize contribution.

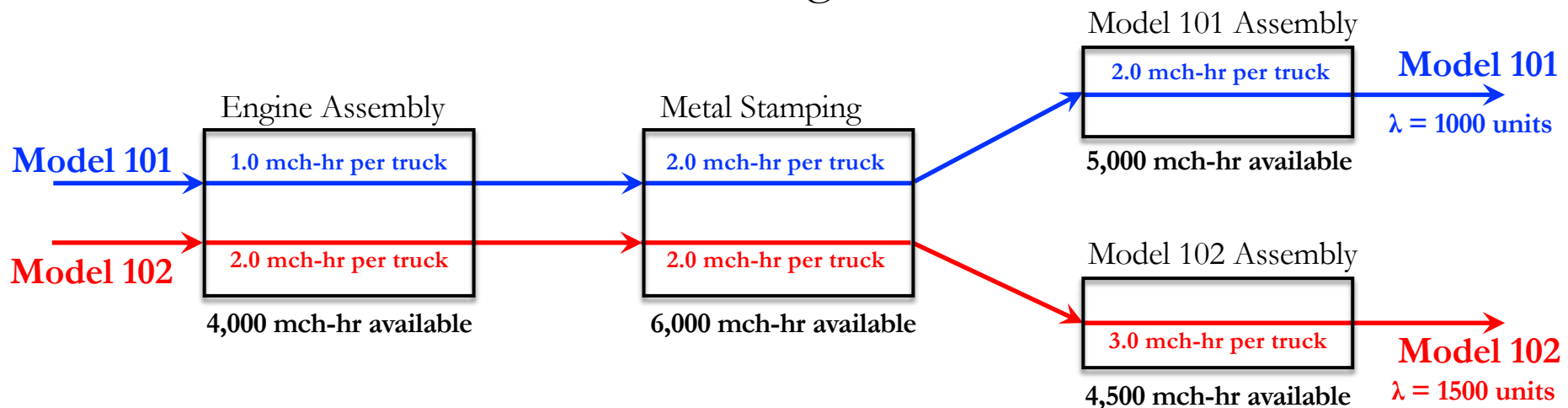
Merton Truck

Contribution towards Fixed Costs is (Revenue – Var. Costs)

- **Model 101:** $39,000 - 24,000 - 4,000 - 8,000 = \$3,000$ per truck
- **Model 102:** $38,000 - 20,000 - 4,500 - 8,500 = \$5,000$ per truck

THEREFORE WE SHOULD MAKE AS MANY 102s AS POSSIBLE! ???

Process Flow Diagram



Merton Truck

Activity-Resource Consumption Matrix			
	Model 101 Model 102		Capacity
Eng. Assembly	1	2	4000
Metal Stamping	2	2	6000
Model 101 Assem.	2	0	5000
Model 102 Assem.	0	3	4500
Contribution	\$3,000	\$5,000	

Production Strategies

Contribution if only produce Model 101:

- Max Production = $\text{Min}\{4000/1, 6000/2, 5000/2\} = 2500$ units.
- Thus, Contribution = $2500 \times \$3000 = \$7,500,000$

Contribution if only produce Model 102:

- Max Production = $\text{Min}\{4000/2, 6000/2, 4500/3\} = 1500$ units.
- Thus, Contribution = $1500 \times \$5000 = \$7,500,000$

Current Policy: 1000 M101's ➔ Contribution = $1000 \times \$3000 + 1500 \times \5000
 1500 M102's = \$10,500,000

Is there a better policy?

- Are they operating at their capacity with this policy?

A Framework for Capacity Planning

Let:

X_1 = Number of 101 to produce

X_2 = Number of 102 to produce

← Decision Variables

We seek to:

Maximize $3000 X_1 + 5000 X_2$

← Objective Function

Subject to:

Engine Assembly: $X_1 + 2X_2 \leq 4000$

Metal Stamping: $2X_1 + 2X_2 \leq 6000$

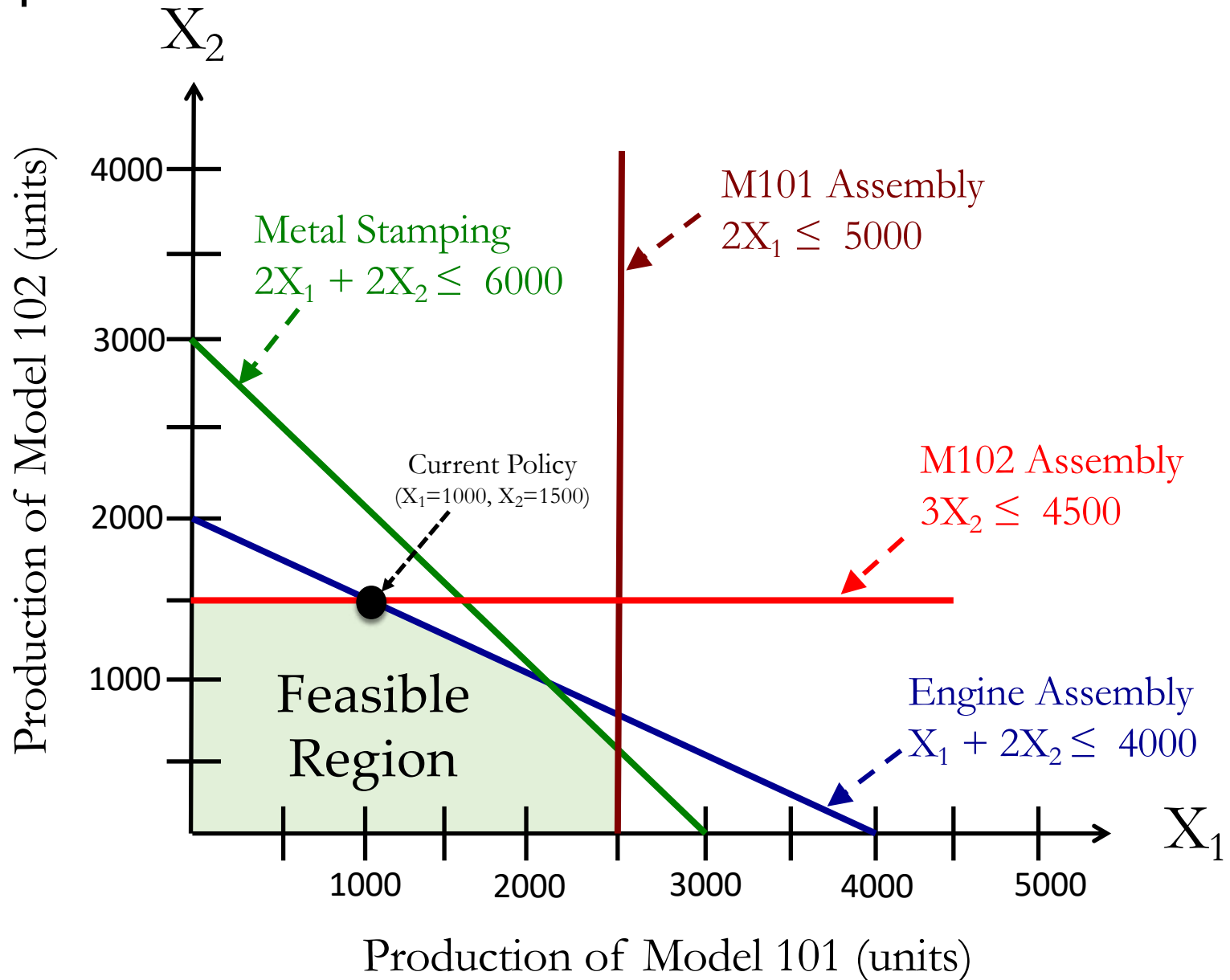
Model 101 Assembly: $2X_1 \leq 5000$

Model 102 Assembly: $3X_2 \leq 4500$

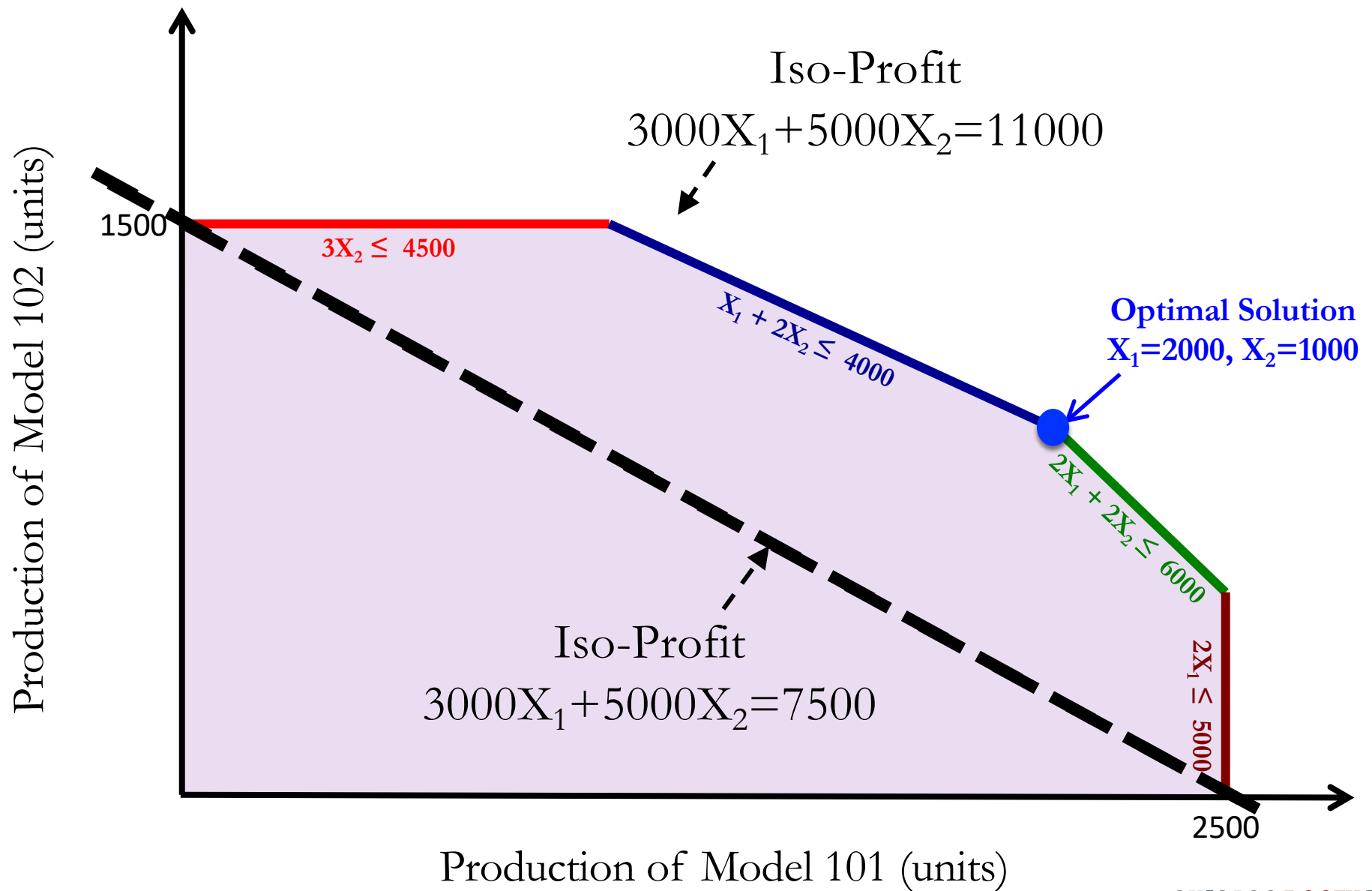
Non-negativity: $X_1 \geq 0, X_2 \geq 0$

} ← Constraints

Graph Constraints



Find Optimal Solution



Excel Solver

- How to solve larger problems
- Which resource(s) should I expand?
- How much would I pay for one more unit of a given resource?
- How sensitive are my strategies to input data?

Excel Solver

Mathematical Format

Maximize $3000 X_1 + 5000 X_2$

Subject to:

Engine Assembly:

$X_1 + 2X_2 \leq 4000$

Metal Stamping:

$2X_1 + 2X_2 \leq 6000$

Model 101 Assembly:

$2X_1 \leq 5000$

Model 102 Assembly:

$3X_2 \leq 4500$

Non-negativity:

$X_1 \geq 0, X_2 \geq 0$

Excel Format

	A	B	C	D	E	F	G
1			MERTON TRUCK COMPANY				
2							
3		Variables	Model 101	Model 102		Total Contribution	
4		Number to Make	0	0			
5		Unit Contribution	3000	5000		\$0	
6		Constraints			Used	Available	
7		Engine Assembly	1	2	0	4000	
8		Metal Stamping	2	2	0	6000	
9		101 Assembly	2	0	0	5000	
10		102 Assembly	0	3	0	4500	
11							
12		Inputs					
13		Outputs					
14							

	A	B	C	D	E	F	G
1			MERTON TRUCK COMPANY				
2							
3		Variables	Model 101	Model 102		Total Contribution	
4		Number to Make	0	0			
5		Unit Contribution	3000	5000		=+SUMPRODUCT(C4:D4,C5:D5)	
6		Constraints				Available	
7		Engine Assembly	1	2	=+SUMPRODUCT(C7:D7,C\$4:D\$4)	4000	
8		Metal Stamping	2	2	=+SUMPRODUCT(C8:D8,C\$4:D\$4)	6000	
9		101 Assembly	2	0	=+SUMPRODUCT(C9:D9,C\$4:D\$4)	5000	
10		102 Assembly	0	3	=+SUMPRODUCT(C10:D10,C\$4:D\$4)	4500	
11							
12		Inputs					
13		Outputs					
14							

Excel Solver (cont'd)

Excel Format

MERTON TRUCK COMPANY					
Variables	Model 101	Model 102	Total Contribution		
Number to Make	0	0			
Unit Contribution	3000	5000	\$0		
Constraints			Used	Available	
Engine Assembly	1	2	0	4000	
Metal Stamping	2	2	0	6000	
101 Assembly	2	0	0	5000	
102 Assembly	0	3	0	4500	
Inputs					
Outputs					

Solver

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Close

Solve

Answer Report

Objective Cell (Max)

Cell	Name	Original Value	Final Value
\$F\$5	Unit Contribution Total Contribution	\$0	\$11,000,000

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$C\$4	Number to Make Model 101	0	2000	Contin
\$D\$4	Number to Make Model 102	0	1000	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
\$E\$7	Engine Assembly	4000	\$E\$7<=\$F\$7	Binding	0
\$E\$8	Metal Stamping	6000	\$E\$8<=\$F\$8	Binding	0
\$E\$9	101 Assembly	4000	\$E\$9<=\$F\$9	Not Binding	1000
\$E\$10	102 Assembly	3000	\$E\$10<=\$F\$10	Not Binding	1500

Shadow Prices

- How much benefit do I get by adding an additional unit of a resource?

Excel Solver Sensitivity Report

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$C\$4	Number to Make Model 101	2000	0	3000	2000	500
\$D\$4	Number to Make Model 102	1000	0	5000	1000	2000

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$E\$7	Engine Assembly Used	4000	2000	4000	500	500
\$E\$8	Metal Stamping Used	6000	500	6000	500	1000
\$E\$9	101 Assembly Used	4000	0	5000	1E+30	1000
\$E\$10	102 Assembly Used	3000	0	4500	1E+30	1500

Shadow Prices: They reflect the *marginal* (instantaneous) increment in the objective function if the corresponding resource is marginally increase.

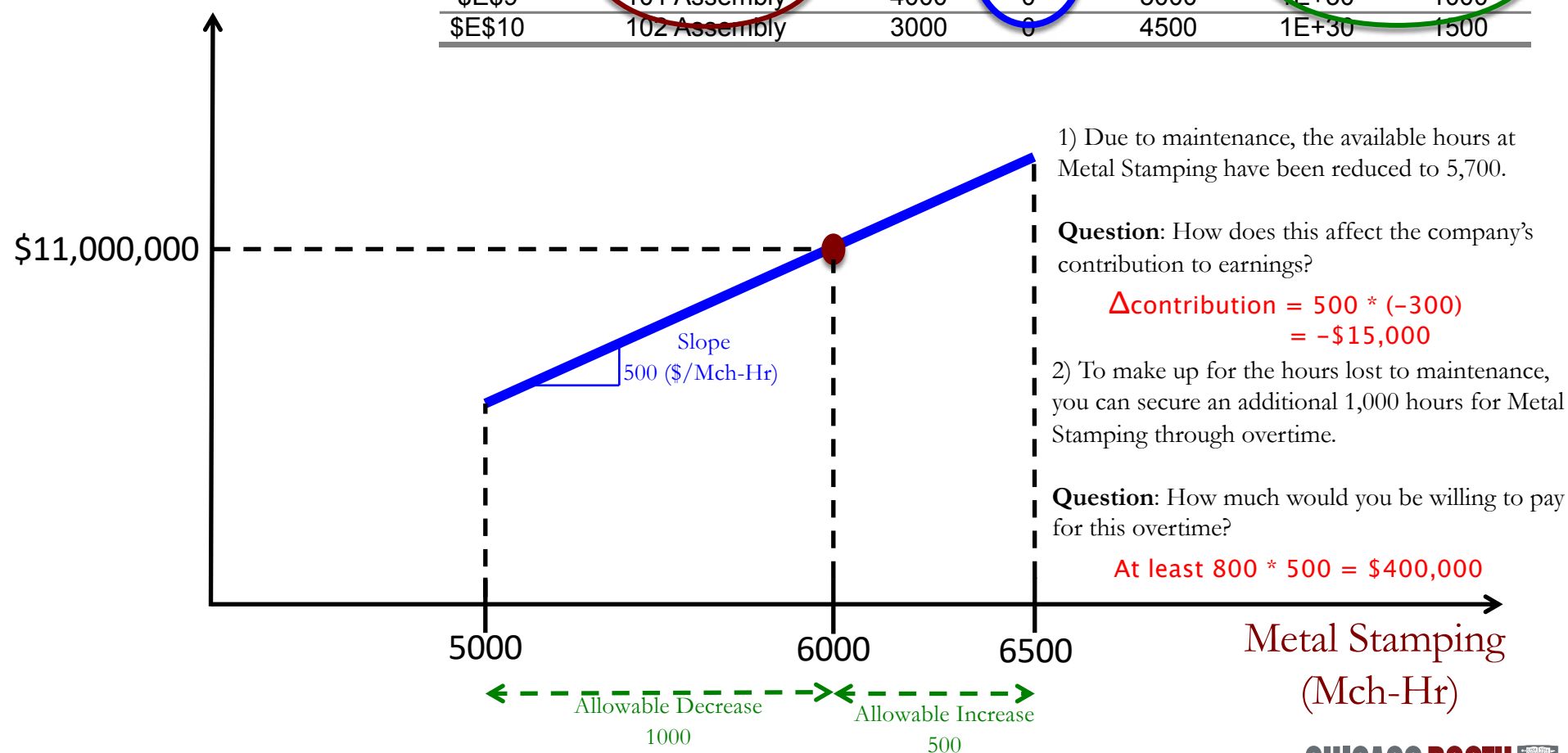
Shadow Prices (cont'd)

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$E\$7	Engine Assembly	4000	2000	4000	500	500
\$E\$8	Metal Stamping	6000	500	6000	500	1000
\$E\$9	101 Assembly	4000	0	5000	1E+30	1000
\$E\$10	102 Assembly	3000	0	4500	1E+30	1500

Contributions

(\$)



Prompt:

The Merton production system manufactures two products: Model 101 and Model 102. The system consists of four activities: Engine Assembly, Metal Stamping, Model 101 Assembly, and Model 102 Assembly. The current available production capacity (in machine hours) for these activities is 4000, 6000, 5000, and 4500 mch-hrs, respectively.

Each unit of Model 101 requires 1 mch-hr of Engine Assembly, 2 mch-hrs of Metal Stamping, and 2 mch-hrs of Model 101 Assembly. Each unit of Model 102 requires 2 mch-hrs of Engine Assembly, 2 mch-hrs of Metal Stamping, and 3 mch-hrs of Model 102 Assembly. The contribution to earnings is \$3000 per unit for Model 101 and \$5000 per unit for Model 102.

Write a Python code that formulates the Merton production problem as a Linear Program, which determines the optimal production levels for Model 101 and Model 102, maximizing total contribution to earnings while adhering to the capacity constraints. It should also produce sensitivity analysis, including shadow prices of the various resources.



```
import numpy as np
from scipy.optimize import linprog

# 1. Define Objective Function Coefficients
# Note: linprog performs minimization, so we negate coefficients to maximize
# Maximize 3000*x1 + 5000*x2 => Minimize -3000*x1 - 5000*x2
c = [-3000, -5000]

# 2. Define Inequality Constraint Matrix (A_ub * x <= b_ub)
# Rows: Engine Assembly, Metal Stamping, Model 101 Assembly, Model 102 Assembly
A = [
    [1, 2], # Engine: 1*x1 + 2*x2 <= 4000
    [2, 2], # Stamping: 2*x1 + 2*x2 <= 6000
    [2, 0], # 101 Assy: 2*x1 + 0*x2 <= 5000
    [0, 3]  # 102 Assy: 0*x1 + 3*x2 <= 4500
]

# 3. Define Right-Hand Side (RHS) of constraints
b = [4000, 6000, 5000, 4500]

# 4. Define Bounds for Decision Variables (x1, x2 >= 0)
x_bounds = (0, None)

# 5. Solve the Linear Program using the 'highs' method (supports dual values)
res = linprog(c, A_ub=A, b_ub=b, bounds=[x_bounds, x_bounds], method='highs')

# 6. Extract Results and Sensitivity Analysis (Shadow Prices)
if res.success:
    print(f"--- Optimal Production Plan ---")
    print(f"Model 101: {res.x[0]:.2f} units")
    print(f"Model 102: {res.x[1]:.2f} units")
    print(f"Total Contribution: ${-res.fun:,.2f}")

    print(f"\n--- Sensitivity Analysis (Shadow Prices) ---")
    # Shadow prices are stored in 'ineqlin.marginals' for the 'highs' method
    resource_names = ["Engine Assembly", "Metal Stamping", "Model 101 Assembly",
                      "Model 102 Assembly"]
    for name, price in zip(resource_names, res.ineqlin.marginals):
        print(f"Shadow Price for {name}: ${price:.2f}")
else:
    print("Optimization failed:", res.message)
```

```
... =====
                        MERTON OPTIMAL PRODUCTION PLAN
=====
Model 101 (x1):      2000.00 units
Model 102 (x2):      1000.00 units
Total Earnings: $11,000,000.00
=====

RESOURCE UTILIZATION & SHADOW PRICES
Resource Capacity Used Shadow Price
Engine Assy      4000 4000.0    -2000.0
Metal Stamp      6000 6000.0    -500.0
M101 Assy        5000 4000.0     -0.0
M102 Assy        4500 3000.0     -0.0

TA Note (BOB):
```

Why the code does not produce the correct sign for the shadow prices?

B BOB – Booth Ops Bot • Custom Gem

Great question! This is a common point of confusion when moving from the **managerial frameworks** we use in class to **computational libraries** like `scipy.optimize`.

As your TA, let me explain the mathematical “why” behind the sign flip and how to reconcile it with Professor Caldentey’s teaching notes.

1. The Solver's Logic: Minimization vs. Maximization

Standard optimization algorithms (including the `highs` solver in Python) are mathematically designed to **minimize** a function. To solve our Merton **maximization** problem, we have to perform a “mathematical trick”:

- Instead of maximizing **Profit**, we minimize **-Profit**.
- Because the solver is working in a “negative world,” the dual variables (shadow prices) it calculates are relative to that negative objective function.

Key Ideas

➤ Capacity and Bottleneck Analysis

- Capacity can be complex, often depending on product mix
- Process flow diagram and build-up graph analysis
- Bottleneck Management
 - Work smarter/Work faster/Work longer /Eliminate waste (lean)
Increase scale (invest) / Increase batch size (economies of scale)

➤ Capacity Expansion Analysis

- What is my path of expansion?
- What are the gains along this path?

➤ LP: A Framework for Optimal Resource Allocation

- LP solver — such as included in Excel
- Marginal Value of a Resource and Shadow Prices
- Effects of a Changing Environment on Operational Decisions

After Class Activities

➤ REVIEW SESSION 1

➤ INDIVIDUAL:

TAKEAWAY #2:

Contribute a second individual idea for final group project, similar to takeaway 1.
(1 paragraph)

➤ GROUP:

CASE: SHOULDICE HOSPITAL

Unique opportunities and challenges in managing service process.